MODULE 17

Survey-Grade Case Study: RIEGL VUX-240[24]

This module is a project-style case study with a survey-grade airborne scanner, the **RIEGL VUX-240[24]** (datasheet dated 2026-08-17), flown on a crewed aircraft, helicopter or large UAS. Every laser and IMU error is pushed to the ground for a **500 ft (152.4 m) return** — conveniently, almost exactly the 150 m range at which RIEGL specifies its accuracy and precision — and the results are compared with an ASPRS accuracy class.

SIMPLE APPROACH

This scanner is like a super-fast, super-careful version of the ones we met before. It fires up to 2.4 million laser flashes a second from an airplane. To know exactly where each dot lands 500 feet below, we have to add up all the little mistakes: the laser's own wobble, the tiny tilt and turn errors of the motion sensor, and where the GPS thinks the airplane is. The surprise: with a scanner this good, the *airplane's position* is the biggest mistake, not the laser.

17.1 The scanner in one table

Specification (RIEGL datasheet)	**VUX-240[24]**
Measuring principle	time of flight; echo digitization; online waveform processing; multiple-target and multiple-time-around (MTA) processing
Pulse repetition rate (PRR)	300, 600, 1200, 1800, 2400 kHz (up to 2.0 M meas./s at 2400 kHz, 75° FOV)
Max. range, ≥20% targets	1200 m (300 kHz) … 460 m (2400 kHz)
Max. operating altitude AGL, ≥20%	2900 ft (300 kHz) … 1100 ft (2400 kHz)
Targets per pulse	up to 30 (300 kHz) … 5 (2400 kHz)
Minimum range	5 m
Accuracy / precision	**20 mm / 15 mm** (one sigma @ 150 m, RIEGL test conditions)
Beam divergence; footprint	0.35 mrad ($1/e^2$); 35 mm @ 100 m, 175 mm @ 500 m
Scanner	rotating polygon mirror, parallel scan lines, FOV ±37.5°, 40–600 lines/s
Angular step / angle resolution	0.002°–0.18° / 0.001°
Timing	GNSS 1PPS input; camera trigger and exposure interfaces (up to 4 cameras)
Laser class	**Class 3R** (IEC 60825-1:2014); NOHD 0 m and ENOHD 0 m when flying faster than 1 kn; stationary: NOHD 0.3 m, ENOHD 3.5 m
Size, weight, power	292×162×185 mm; approx. 4.3 kg; typ. 77 W

IMU/GNSS (post-processed)	RiLOC-F	Applanix AP+30	AP+50	AP+60
Roll, pitch	0.005°	0.010°	0.005°	0.0025°*
Heading	0.005°	0.025°	0.010°	0.005°
IMU sampling	up to 700 Hz	200 Hz	200 Hz	200 Hz
Position (typ.)	0.02–0.03 m	0.02–0.05 m	0.02–0.05 m	0.02–0.05 m
in feet	0.07–0.10 ft	0.07–0.16 ft	0.07–0.16 ft	0.07–0.16 ft

*May require a local gravity model to achieve full accuracy (RIEGL datasheet).

17.2 The project

Client need	corridor topography for a highway widening; hard-surface vertical $\text{RMSE}_V \leq 5$ cm and horizontal $\text{RMSE}_H \leq 5$ cm (ASPRS Edition 2 classes), checkpoints ≤ 2.5 cm
Platform	helicopter or large UAS, 500 ft AGL, 60 kn (30.9 m/s); UAS alternative 10 m/s
Scanner settings	2400 kHz PRR, 75° FOV
Trajectory	IMU/GNSS post-processed (PPK) against a nearby CORS or base; RiLOC-F as the baseline option
Question	which laser and IMU errors matter, how big are they at 500 ft, and does the system meet the class?

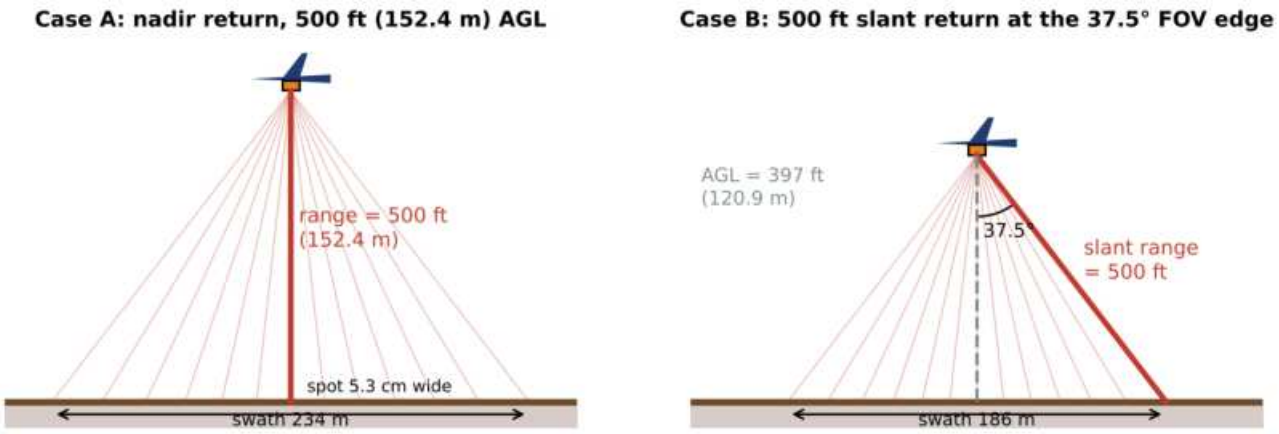


Figure 17.1: The two returns we analyze: Case A straight down from 500 ft AGL; Case B a 500 ft slant return at the edge of the 75° field of view (aircraft at 397 ft AGL).

17.3 Laser errors to consider

RIEGL states accuracy and precision in **millimeters**: 20 mm and 15 mm (one sigma at 150 m). The rest of this module works in **centimeters** so that laser, IMU and GNSS terms can be added on one scale, and gives feet for U.S. practice. The conversions are simply 10 mm = 1 cm and 1 ft = 304.8 mm, so 20 mm = 2.0 cm = 0.066 ft and 15 mm = 1.5 cm = 0.049 ft. Because RIEGL specifies the values at 150 m and our return is at 152.4 m (500 ft), they apply essentially unchanged.

Laser error source	Why it matters	At 500 ft (152.4 m)
Range accuracy (bias)	systematic offset along the beam; removed only by calibration	20 mm = 2.0 cm (0.066 ft)
Range precision (noise)	random scatter; averages down over surfaces	15 mm = 1.5 cm (0.049 ft)
Combined range error	$\sqrt{20^2 + 15^2} = \sqrt{625}$	**25 mm = 2.5 cm (0.082 ft)**
Beam footprint	edge location and mixed returns on curbs, rails, wires	53 mm (5.3 cm) spot $\Rightarrow \pm 27$ mm edge uncertainty
Angle encoder	0.001° resolution $\Rightarrow$ quantization $0.001°/\sqrt{12}$	0.8 mm = 0.08 cm — negligible
Polygon mirror facets	each facet needs its own angular calibration; a bad facet gives repeating stripes	checked in strip comparisons
Range walk / incidence	strong or grazing echoes shift timing slightly; waveform processing reduces it	sub-cm to cm; check on targets
MTA ambiguity	at high PRR several pulses are in the air; wrong zone $\Rightarrow$ gross blunder	see below
Atmosphere	refractive index changes range by ~300 ppm if uncorrected	46 mm = 4.6 cm at 500 ft if ignored!

WORKED EXAMPLE — multiple-time-around (MTA) at 2400 kHz

Unambiguous range

$c/(2\,f_{\mathrm{PRR}}) = 2.997 \times 10^8/(2 \times 2.4 \times 10^6) = 62.4$ m (Module 3)

Which zone?

$152.4/62.4 = 2.44 \Rightarrow$ zone 3: two newer pulses leave before this echo returns

Result: the processor must assign every echo to the correct zone. A wrong assignment moves a point by a whole multiple of 62.4 m — obvious "ghost layers" that are blunders, not noise. At 300 kHz the zone is 500 m deep and the problem disappears at this altitude.

WORKED EXAMPLE — atmospheric correction

Refractive index of air

$n \approx 1.0003$, i.e. 300 ppm

At 500 ft

$152.4 \times 300 \times 10^{-6} = 0.046$ m

Result: 4.6 cm — nearly twice the laser's whole error budget. Processing software applies the correction automatically from temperature and pressure; confirm it is on and the values are sensible (Module 3).

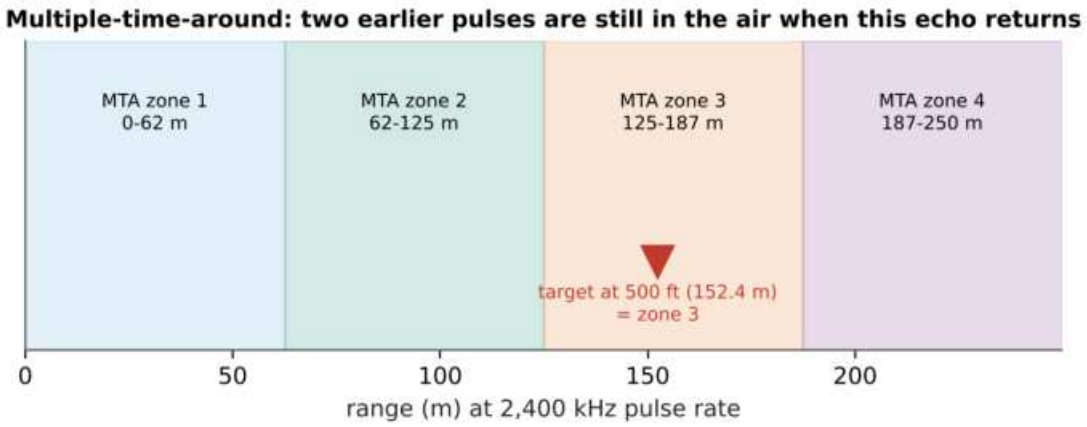


Figure 17.2: At 2.4 MHz the 500 ft echo belongs to the third pulse back.

17.4 IMU/GNSS errors and the 500 ft error budget

The post-processed IMU/GNSS options (table above) are carried through a full error budget for a 500 ft return in Module 14 (Section 14.11.3). Summary (1σ, before ground control): RiLOC-F 3.7–3.9 cm H and 4.4–4.5 cm V (0.12–0.13 ft H, 0.15 ft V); AP+30 5.5–6.6 cm H and 5.8–5.9 cm V (0.18–0.22 ft H, 0.19 ft V). After ground control and strip adjustment the RiLOC-F reaches about 3.1 cm H and 3.1 cm V (0.10 ft), supporting a 5 cm class with margin. GNSS position — not the laser — is the largest term.

17.5 Point density at 500 ft

WORKED EXAMPLE — point density at 500 ft

Swath

$2 \times 152.4 \times \tan 37.5^\circ = 234$ m

At 60 kn (30.9 m/s), 2.0 M meas./s

$2.0 \times 10^6 / (234 \times 30.9) = 277$ points/m^2

UAS at 10 m/s

$2.0 \times 10^6 / (234 \times 10) = 855$ points/m^2

Result: consistent with RIEGL's example of ~126 pts/m^2 at 1,100 ft and 60 kn (density scales with 1/altitude). At 277 pts/m^2 points are about 6 cm apart — similar to the 5.3 cm footprint, so features are sampled about once per spot.

17.6 Survey-grade checklist for this system

Laser	range calibration current; atmospheric correction on; MTA zones resolved (no ghost layers); facet calibration checked; minimum range 5 m respected
IMU/GNSS	post-processed (PPK) trajectory with short baselines; IMU option matched to the class (heading drives swath-edge horizontal error); gravity model if using AP+60
Calibration	lever arm measured to mm; boresight and facet calibration flown over a calibration site with opposite and cross strips
Timing	1PPS connected and logged; camera exposure events time-stamped
Control	ground control for strip adjustment; independent checkpoints ($\geq$30), 2$\times$ more accurate than the product (Module 13)
Safety	Class 3R: safe from the air (>1 kn), but on the ground the NOHD is 0.3 m and ENOHD 3.5 m — never look into the aperture or test-fire toward people
Reporting	ASPRS Edition 2 statement; swath-to-swath and within-swath precision results; client accuracy agreement (Appendix E)

BIG IDEA

At 500 ft, a survey-grade laser is no longer the weak link: the RIEGL's own range error (2.5 cm) is comparable to the GNSS error, and the IMU's attitude error is 1–7 cm depending on the option. Survey-grade results come from the *whole system* — trajectory, calibration, control and checkpoints — not from the laser alone.

TRY IT YOURSELF

17.1 What ground displacement does a 0.010° heading error cause at the swath edge for a 500 ft slant return?
17.2 At 1200 kHz PRR, which MTA zone does a 500 ft echo fall in?
17.3 If GNSS vertical error could be cut to 1.0 cm, what is the RiLOC-F nadir vertical total (range 2.5 cm, lever 0.5 cm)?
17.4 (Simple) Why is the RIEGL safe to fly over people but not safe to stare into on the ground?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," of the Reference Book, starting on page 163.

SOURCES FOR THIS MODULE

RIEGL, *VUX-240*[24] *datasheet* (2026-08-17), including RiLOC-F and Applanix AP+30/50/60 IMU/GNSS options; Glennie (2007) for airborne error geometry; Module 3 for MTA ranging.